T2-22 parameter section 工程计算书

本计算书记录 AC2 · long random report · 随机响应 · random 的空间刚架随机响应模型、输入参数与结果，供计算结果复核。零均值平稳随机响应；RMS 为统计均方根，不是瞬时峰值或确定性位移；输入有效频段外为零

1 模型概况与来源

项目 T2-22 parameter section；项目 ID project_be5ee344-7eb4-4af0-a922-b39feb9b3a8c

分析 ID analysis_d2b0b23a-117a-4217-a76d-d057b5cef5cf；输入修订 R13；状态 已完成；收敛状态 converged；历史结果 已过期

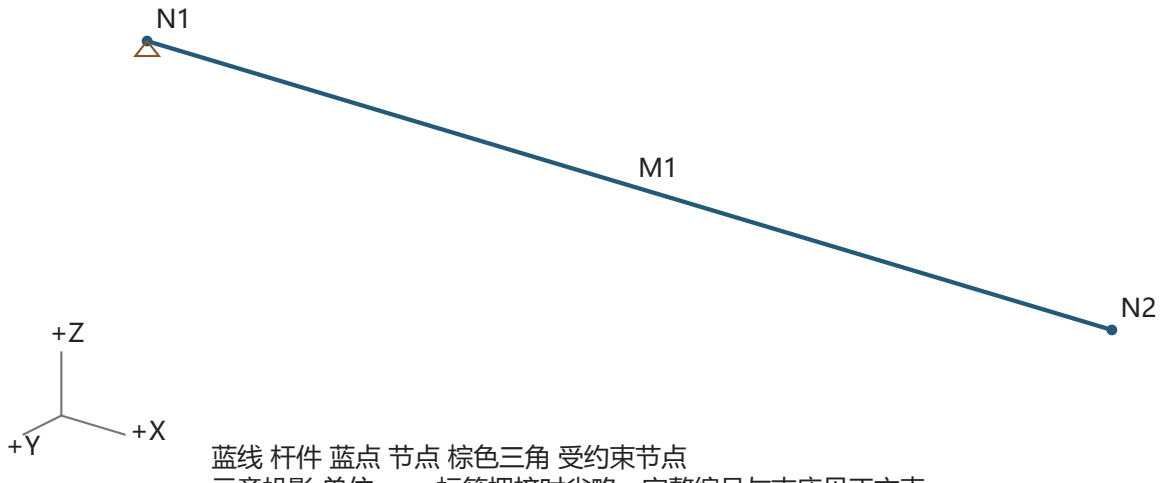
分析时间 2026-10-07T06:42:06Z；报告生成时间 2026-10-07T06:48:47.616Z（ISO 8601）

运行来源 formal-model；请求 request_8fe65170-a53f-467b-8117-631ff78b93ca；实际阶段 模型检查 (2026-10-07T06:42:06Z) → 刚度组装 (2026-10-07T06:42:06Z) → 方程求解 (2026-10-07T06:42:06Z) → 结果恢复 (2026-10-07T06:42:06Z)。

选定来源 AC2 · long random report · 随机响应 · random；2 个节点，1 根杆件，1 个支座。正文、表格及图片均使用本次分析的冻结输入与输出。

显示单位：坐标与长度 mm；力 N；力矩 N·mm；位移 mm；转角 rad；均布荷载 N/mm；E MPa；密度 kg/mm³；面积 mm²；Iy、Iz、J mm⁴。内部规范单位及冻结数值未改变。

冻结输入模型
AC2 · long random report · 随机响应 · random · R13



蓝线 杆件 蓝点 节点 棕色三角 受约束节点
示意投影 单位 mm 标签拥挤时省略；完整编号与支座见正文表

分析 analysis_d2b0b23a-117a-4217-a76d-d057b5cef5cf

图 1 冻结输入模型。全局 +X、+Y、+Z 方向见图，杆件连接和支座自由度以正文表为准。大模型省略对象标签，稳定 ID 见表。

2 理论假设与适用边界

空间 Euler–Bernoulli 直杆单元，每节点 ux、uy、uz、rx、ry、rz 六个自由度。线弹性、小变形，未释放分量为刚接，忽略剪切变形；支座采用零位移自由度约束。

模态叠加频响 H(f)，ω = 2πf；单源 Sresponse = |H|² Sinput，同源空间作用先在复频响中相加；异源独立，响应自谱相加。RMS = √∫ Sresponse(f) df，采用保存频点的梯形积分。

整体一致平动基底以 −M r a 输入，输出为相对基底的平移响应；外力输出为固定全局参考的平移响应。仅支持所选工程节点的一个全局平移分量，不包含绝对加速度、转角、内力或峰值。

零均值平稳随机响应，RMS 是统计均方根。本报告不推导峰值、安全或规范合格结论。

总体位移、转角、支座反力及反力矩按全局 XYZ。局部 x 由杆件起点指向终点，y、z 随截面转角；局部轴向量见表。杆端 N、Vy、Vz、T、My、Mz 为局部轴端部作用分量，两个端点分别记录，不能直接当作截面内力包络。

求解器 framemind-native/8；算法 generalized-self-adjoint；约束 free-dof-elimination；相对主元阈值 1e-12。

3 材料截面与模型参数

材料参数

材料	名称	E (MPa)	ν	ρ (kg/mm³)
M1	E200	200000	0.3	0.00000785

M1 · material_0001：用户自定义参数

截面参数

截面	名称	A (mm²)	Iy (mm⁴)	Iz (mm⁴)	J (mm⁴)
S1	rectangle	20000	66666670	16666670	45776040

S1 · section_0001：用户自定义参数

S1 · 实心矩形；高度 h（局部 z） = 200 mm；宽度 b（局部 y） = 100 mm。idealized-section-v1；SCI P385 (2011), Appendix B.2; A/I by area integration；https://www.steelconstruction.info/images/6/6f/Sci_p385.pdf。J：SCI B.2.1 矩形近似；A/I 为理想矩形精确积分。

节点坐标与稳定标识

节点	ID	X (mm)	Y (mm)	Z (mm)
N1	node_0001	0	0	0
N2	node_0002	3000	0	0

杆件连接、分配与截面朝向

杆件及 ID	起点 → 终点	材料 / 截面	长度 (mm)	转角 (°)
M1 member_0001	N1 → N2	M1 / S1	3000	0

Member / ID	起点局部释放	终点局部释放
M1 / member_0001	刚接	刚接

局部单位轴向量按全局 XYZ 分量表示

杆件	局部 x	局部 y	局部 z
M1	1, 0, 0	0, 1, 0	0, 0, 1

4 支座与边界条件

约束表示该自由度位移或转角为零；自由表示未施加该约束。平移方向为全局 XYZ，转动方向为绕全局 XYZ。

支座 / 节点	ux	uy	uz	rx	ry	rz
B1 / N1	约束	约束	约束	约束	约束	约束

5 随机激励与复现设置

所有作用方向按全局 XYZ。同源共享一个随机过程，空间作用系数含符号；不同源相互独立，不含互谱或任意相关性。输入为单边 Hz 谱，线性插值，有效频段外为零。

源 ID / 输入谱 ID	激励类型 / 工程对象 ID	全局方向 XYZ	有符号系数 / 单位
force / PSD2	节点力 / node_0002	0, 1, 0	1 / 无量纲

完整频段 0 — 25 Hz；初始频点 257；最终频点 2193；频点预算 20001；最大加密 10；积分 trapezoidal（梯形）；响应 PSD 绝对容差 1e-12 mm²/Hz；方差绝对容差 1e-12 mm²；RMS 绝对容差 1e-9 mm。

冻结设置字段	值（内部规范单位 m、Hz、m²/Hz；方差 m²）
absolutePsdTolerance	1e-18
absoluteRmsTolerance	1e-12
absoluteVarianceTolerance	1e-18
frequencyPoints	257
integration	trapezoidal
maxFrequencyPoints	20001
maxRefinements	10
modalAnalysisId	analysis_c52c386c-0e16-4663-aedd-a0feac3f46b4
modeIncrement	2
relativeTolerance	0.005
sourceRelationship	independent

冻结模态基础、质量与阻尼

模态分析 analysis_c52c386c-0e16-4663-aedd-a0feac3f46b4；输入修订 R8；模态时间 2026-10-07T05:02:02Z；求解器 framemind-native/5；实际采用 6 阶；模态网格 16 单元/杆；网格收敛 converged。

结构质量 471 kg；附加质量 0 kg；总质量 471 kg。质量不重复作为静力自重输入。振型按单位广义质量归一化，幅值不是确定性位移。

工程杆件 ID	材料 / 截面 ID	ρ (kg/mm³)	质量 (kg)
member_0001	material_0001 / section_0001	0.00000785	471

附加质量来源	工程对象 ID	质量 (kg)	线质量 (kg/mm)

模态设置字段	内部规范值
frequencyTolerance	0.002
maxDivisions	16
maxEquations	600
modeCount	6

阻尼绑定 analysis_c52c386c-0e16-4663-aedd-a0feac3f46b4；2026-10-07T05:02:02Z；统一阻尼比 0.02；来源 user；依据 Independent analytical reference。覆盖频率 25 Hz；增阶步长 2；模态预算 50。

阶次 / 是否采用	频率 (Hz)	阻尼比 / 覆盖	来源 / 依据
1 / 采用	9.059787	0.02 / 统一	user / Independent analytical reference
2 / 采用	18.11957	0.02 / 统一	user / Independent analytical reference
3 / 采用	56.777	0.02 / 统一	user / Independent analytical reference
4 / 采用	113.554	0.02 / 统一	user / Independent analytical reference
5 / 采用	158.9827	0.02 / 统一	user / Independent analytical reference
6 / 采用	193.4178	0.02 / 统一	user / Independent analytical reference

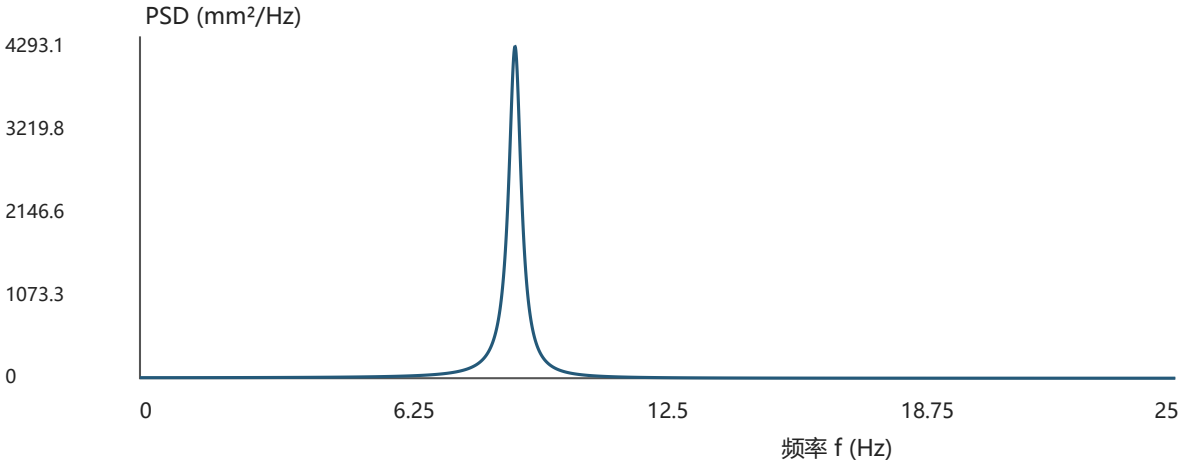
6 响应 PSD、RMS 与收敛

工程节点 N2 / node_0002；平移分量 Y；坐标系 global XYZ；参考系 固定全局参考；单边 PSD，Hz。

RMS 统计均方根	方差	PSD 单位	参考系
49.43695 mm	2444.012 mm²	mm²/Hz	固定全局参考

随机平移响应 PSD（完整保存频点）

AC2 · long random report · 随机响应 · random · R13



node_0002 / Y 全局 XYZ 固定全局参考 单边谱

分析 analysis_d2b0b23a-117a-4217-a76d-d057b5cef5cf

PSD 图使用全部 2193 个保存频点，无抽样或重新求解。RMS 使用整个保存频段，不以正文截取表积分。

频点数	方差 (mm²)	RMS (mm)	PSD / 方差 / RMS 误差比
275	2447.087	49.46804	基线 / 基线 / 基线
549	2444.658	49.44348	0.03357784 / 0.000992906 / 0.0004965763
1097	2444.138	49.43822	0.01126142 / 0.0002126122 / 0.0001063117
2193	2444.012	49.43695	0.003486093 / 0.00005152687 / 0.00002576377

模态截断 converged；对比 4 → 6 阶；PSD 误差比 0.0001335571；RMS 误差比 0.00003772695；相对容差 0.005。

7 频点表与完整数据入口

正文数值保留七位有效数字。已选择完整频点附录，表内列出全部输入谱和响应频点。 工程计算书窗口的“导出随机数据 ZIP”提供 manifest.json、analysis.json（完整冻结输入、模态基础和结果）、response.csv、rms.csv、input-N.csv 与 process.json；CSV 保留完整数值精度及内部 / 显示单位两列。

响应 PSD：完整 2193 点

原表索引	频率 (Hz)	PSD (mm²/Hz)
1	0	7.26898
2	0.01220703	7.269005
3	0.02441406	7.269082

原表索引	频率 (Hz)	PSD (mm²/Hz)
4	0.03662109	7.26921
5	0.04882813	7.26939
6	0.06103516	7.269621
7	0.07324219	7.269903
8	0.08544922	7.270237
9	0.09765625	7.270622
10	0.1098633	7.271058
11	0.1220703	7.271545
12	0.1342773	7.272084
13	0.1464844	7.272675
14	0.1586914	7.273317
15	0.1708984	7.27401
16	0.1831055	7.274755
17	0.1953125	7.275551
18	0.2075195	7.276399
19	0.2197266	7.277298
20	0.2319336	7.278249
21	0.2441406	7.279252
22	0.2563477	7.280306
23	0.2685547	7.281411
24	0.2807617	7.282569
25	0.2929688	7.283778
26	0.3051758	7.285039
27	0.3173828	7.286352
28	0.3295898	7.287717
29	0.3417969	7.289133
30	0.3540039	7.290602
31	0.3662109	7.292123
32	0.378418	7.293695
33	0.390625	7.29532
34	0.402832	7.296997
35	0.4150391	7.298726

原表索引	频率 (Hz)	PSD (mm²/Hz)
36	0.4272461	7.300508
37	0.4394531	7.302341
38	0.4516602	7.304227
39	0.4638672	7.306166
40	0.4760742	7.308157
41	0.4882813	7.310201
42	0.5004883	7.312297
43	0.5126953	7.314446
44	0.5249023	7.316648
45	0.5371094	7.318903
46	0.5493164	7.32121
47	0.5615234	7.323571
48	0.5737305	7.325984
49	0.5859375	7.328451
50	0.5981445	7.330971
51	0.6103516	7.333545
52	0.6225586	7.336172
53	0.6347656	7.338852
54	0.6469727	7.341586
55	0.6591797	7.344373
56	0.6713867	7.347215
57	0.6835938	7.35011
58	0.6958008	7.353059
59	0.7080078	7.356062
60	0.7202148	7.359119
61	0.7324219	7.362231
62	0.7446289	7.365397
63	0.7568359	7.368617
64	0.769043	7.371892
65	0.78125	7.375222
66	0.793457	7.378606
67	0.8056641	7.382046

原表索引	频率 (Hz)	PSD (mm²/Hz)
68	0.8178711	7.38554
69	0.8300781	7.38909
70	0.8422852	7.392694
71	0.8544922	7.396354
72	0.8666992	7.40007
73	0.8789063	7.403841
74	0.8911133	7.407668
75	0.9033203	7.411551
76	0.9155273	7.41549
77	0.9277344	7.419485
78	0.9399414	7.423536
79	0.9521484	7.427644
80	0.9643555	7.431808
81	0.9765625	7.436029
82	0.9887695	7.440306
83	1.000977	7.444641
84	1.013184	7.449032
85	1.025391	7.453481
86	1.037598	7.457988
87	1.049805	7.462552
88	1.062012	7.467173
89	1.074219	7.471853
90	1.086426	7.476591
91	1.098633	7.481386
92	1.11084	7.48624
93	1.123047	7.491153
94	1.135254	7.496125
95	1.147461	7.501155
96	1.159668	7.506244
97	1.171875	7.511392
98	1.184082	7.5166
99	1.196289	7.521868

原表索引	频率 (Hz)	PSD (mm²/Hz)
100	1.208496	7.527195
101	1.220703	7.532582
102	1.23291	7.538029
103	1.245117	7.543537
104	1.257324	7.549105
105	1.269531	7.554733
106	1.281738	7.560423
107	1.293945	7.566174
108	1.306152	7.571986
109	1.318359	7.577859
110	1.330566	7.583794
111	1.342773	7.589791
112	1.35498	7.59585
113	1.367188	7.601972
114	1.379395	7.608156
115	1.391602	7.614402
116	1.403809	7.620712
117	1.416016	7.627085
118	1.428223	7.633521
119	1.44043	7.640021
120	1.452637	7.646585
121	1.464844	7.653213
122	1.477051	7.659905
123	1.489258	7.666662
124	1.501465	7.673484
125	1.513672	7.680371
126	1.525879	7.687323
127	1.538086	7.69434
128	1.550293	7.701424
129	1.5625	7.708573
130	1.574707	7.715789
131	1.586914	7.723071

原表索引	频率 (Hz)	PSD (mm²/Hz)
132	1.599121	7.730421
133	1.611328	7.737837
134	1.623535	7.745321
135	1.635742	7.752872
136	1.647949	7.760492
137	1.660156	7.768179
138	1.672363	7.775935
139	1.68457	7.78376
140	1.696777	7.791654
141	1.708984	7.799617
142	1.721191	7.80765
143	1.733398	7.815753
144	1.745605	7.823926
145	1.757813	7.832169
146	1.77002	7.840484
147	1.782227	7.848869
148	1.794434	7.857326
149	1.806641	7.865855
150	1.818848	7.874456

原表索引	频率 (Hz)	PSD (mm²/Hz)
151	1.831055	7.883129
152	1.843262	7.891875
153	1.855469	7.900695
154	1.867676	7.909587
155	1.879883	7.918553
156	1.89209	7.927594
157	1.904297	7.936709
158	1.916504	7.945898
159	1.928711	7.955163
160	1.940918	7.964503
161	1.953125	7.973919
162	1.965332	7.983411

原表索引	频率 (Hz)	PSD (mm²/Hz)
163	1.977539	7.99298
164	1.989746	8.002625
165	2.001953	8.012348
166	2.01416	8.022149
167	2.026367	8.032027
168	2.038574	8.041984
169	2.050781	8.05202
170	2.062988	8.062135
171	2.075195	8.07233
172	2.087402	8.082604
173	2.099609	8.092959
174	2.111816	8.103395
175	2.124023	8.113912
176	2.13623	8.124511
177	2.148438	8.135192
178	2.160645	8.145955
179	2.172852	8.156801
180	2.185059	8.16773
181	2.197266	8.178743
182	2.209473	8.189841
183	2.22168	8.201023
184	2.233887	8.21229
185	2.246094	8.223643
186	2.258301	8.235081
187	2.270508	8.246606
188	2.282715	8.258218
189	2.294922	8.269918
190	2.307129	8.281705
191	2.319336	8.293581
192	2.331543	8.305546
193	2.34375	8.3176
194	2.355957	8.329744

原表索引	频率 (Hz)	PSD (mm²/Hz)
195	2.368164	8.341978
196	2.380371	8.354303
197	2.392578	8.366719
198	2.404785	8.379228
199	2.416992	8.391828
200	2.429199	8.404522
201	2.441406	8.417309
202	2.453613	8.43019
203	2.46582	8.443166
204	2.478027	8.456237
205	2.490234	8.469403
206	2.502441	8.482666
207	2.514648	8.496025
208	2.526855	8.509482
209	2.539063	8.523036
210	2.55127	8.53669
211	2.563477	8.550442
212	2.575684	8.564294
213	2.587891	8.578246
214	2.600098	8.592299
215	2.612305	8.606454
216	2.624512	8.62071
217	2.636719	8.63507
218	2.648926	8.649533
219	2.661133	8.6641
220	2.67334	8.678771
221	2.685547	8.693548
222	2.697754	8.708431
223	2.709961	8.72342
224	2.722168	8.738517
225	2.734375	8.753721
226	2.746582	8.769035

原表索引	频率 (Hz)	PSD (mm²/Hz)
227	2.758789	8.784458
228	2.770996	8.79999
229	2.783203	8.815634
230	2.79541	8.831389
231	2.807617	8.847256
232	2.819824	8.863236
233	2.832031	8.87933
234	2.844238	8.895538
235	2.856445	8.911861
236	2.868652	8.9283
237	2.880859	8.944856
238	2.893066	8.961529
239	2.905273	8.97832
240	2.91748	8.995231
241	2.929688	9.012261
242	2.941895	9.029411
243	2.954102	9.046684
244	2.966309	9.064078
245	2.978516	9.081595
246	2.990723	9.099237
247	3.00293	9.117003
248	3.015137	9.134894
249	3.027344	9.152912
250	3.039551	9.171058
251	3.051758	9.189331
252	3.063965	9.207734
253	3.076172	9.226267
254	3.088379	9.244931
255	3.100586	9.263726
256	3.112793	9.282654
257	3.125	9.301716
258	3.137207	9.320913

原表索引	频率 (Hz)	PSD (mm²/Hz)
259	3.149414	9.340245
260	3.161621	9.359714
261	3.173828	9.37932
262	3.186035	9.399065
263	3.198242	9.418949
264	3.210449	9.438974
265	3.222656	9.459141
266	3.234863	9.47945
267	3.24707	9.499903
268	3.259277	9.5205
269	3.271484	9.541243
270	3.283691	9.562133
271	3.295898	9.583171
272	3.308105	9.604359
273	3.320313	9.625696
274	3.33252	9.647184
275	3.344727	9.668825
276	3.356934	9.690619
277	3.369141	9.712568
278	3.381348	9.734673
279	3.393555	9.756935
280	3.405762	9.779356
281	3.417969	9.801935
282	3.430176	9.824676
283	3.442383	9.847578
284	3.45459	9.870644
285	3.466797	9.893874
286	3.479004	9.917269
287	3.491211	9.940832
288	3.503418	9.964563
289	3.515625	9.988464
290	3.527832	10.01254

原表索引	频率 (Hz)	PSD (mm²/Hz)
291	3.540039	10.03678
292	3.552246	10.0612
293	3.564453	10.08579
294	3.57666	10.11056
295	3.588867	10.13551
296	3.601074	10.16063
297	3.613281	10.18594
298	3.625488	10.21143
299	3.637695	10.2371
300	3.649902	10.26296

原表索引	频率 (Hz)	PSD (mm²/Hz)
301	3.662109	10.289
302	3.674316	10.31524
303	3.686523	10.34166
304	3.69873	10.36827
305	3.710938	10.39508
306	3.723145	10.42208
307	3.735352	10.44928
308	3.747559	10.47667
309	3.759766	10.50427
310	3.771973	10.53206
311	3.78418	10.56006
312	3.796387	10.58827
313	3.808594	10.61667
314	3.820801	10.64529
315	3.833008	10.67412
316	3.845215	10.70316
317	3.857422	10.73241
318	3.869629	10.76187
319	3.881836	10.79156
320	3.894043	10.82146
321	3.90625	10.85158

原表索引	频率 (Hz)	PSD (mm²/Hz)
322	3.918457	10.88193
323	3.930664	10.9125
324	3.942871	10.94329
325	3.955078	10.97432
326	3.967285	11.00557
327	3.979492	11.03706
328	3.991699	11.06878
329	4.003906	11.10074
330	4.016113	11.13294
331	4.02832	11.16538
332	4.040527	11.19806
333	4.052734	11.23099
334	4.064941	11.26416
335	4.077148	11.29759
336	4.089355	11.33127
337	4.101563	11.3652
338	4.11377	11.39938
339	4.125977	11.43383
340	4.138184	11.46854
341	4.150391	11.50351
342	4.162598	11.53875
343	4.174805	11.57425
344	4.187012	11.61003
345	4.199219	11.64608
346	4.211426	11.68241
347	4.223633	11.71902
348	4.23584	11.7559
349	4.248047	11.79307
350	4.260254	11.83053
351	4.272461	11.86828
352	4.284668	11.90632
353	4.296875	11.94465

原表索引	频率 (Hz)	PSD (mm²/Hz)
354	4.309082	11.98329
355	4.321289	12.02222
356	4.333496	12.06146
357	4.345703	12.101
358	4.35791	12.14085
359	4.370117	12.18102
360	4.382324	12.2215
361	4.394531	12.2623
362	4.406738	12.30342
363	4.418945	12.34487
364	4.431152	12.38664
365	4.443359	12.42874
366	4.455566	12.47118
367	4.467773	12.51396
368	4.47998	12.55708
369	4.492188	12.60055
370	4.504395	12.64436
371	4.516602	12.68852
372	4.528809	12.73304
373	4.541016	12.77792
374	4.553223	12.82316
375	4.56543	12.86877
376	4.577637	12.91475
377	4.589844	12.9611
378	4.602051	13.00782
379	4.614258	13.05493
380	4.626465	13.10243
381	4.638672	13.15032
382	4.650879	13.19859
383	4.663086	13.24727
384	4.675293	13.29635
385	4.6875	13.34583

原表索引	频率 (Hz)	PSD (mm²/Hz)
386	4.699707	13.39573
387	4.711914	13.44604
388	4.724121	13.49677
389	4.736328	13.54792
390	4.748535	13.5995
391	4.760742	13.65151
392	4.772949	13.70396
393	4.785156	13.75686
394	4.797363	13.8102
395	4.80957	13.86399
396	4.821777	13.91824
397	4.833984	13.97295
398	4.846191	14.02812
399	4.858398	14.08377
400	4.870605	14.1399
401	4.882813	14.1965
402	4.89502	14.2536
403	4.907227	14.31119
404	4.919434	14.36928
405	4.931641	14.42787
406	4.943848	14.48697
407	4.956055	14.54659
408	4.968262	14.60673
409	4.980469	14.6674
410	4.992676	14.7286
411	5.004883	14.79034
412	5.01709	14.85262
413	5.029297	14.91546
414	5.041504	14.97886
415	5.053711	15.04282
416	5.065918	15.10736
417	5.078125	15.17247

原表索引	频率 (Hz)	PSD (mm²/Hz)
418	5.090332	15.23816
419	5.102539	15.30445
420	5.114746	15.37134
421	5.126953	15.43884
422	5.13916	15.50694
423	5.151367	15.57567
424	5.163574	15.64503
425	5.175781	15.71502
426	5.187988	15.78565
427	5.200195	15.85693
428	5.212402	15.92888
429	5.224609	16.00149
430	5.236816	16.07477
431	5.249023	16.14874
432	5.26123	16.2234
433	5.273438	16.29876
434	5.285645	16.37483
435	5.297852	16.45162
436	5.310059	16.52913
437	5.322266	16.60738
438	5.334473	16.68637
439	5.34668	16.76612
440	5.358887	16.84663
441	5.371094	16.92791
442	5.383301	17.00998
443	5.395508	17.09284
444	5.407715	17.1765
445	5.419922	17.26098
446	5.432129	17.34628
447	5.444336	17.43241
448	5.456543	17.51939
449	5.46875	17.60723

原表索引	频率 (Hz)	PSD (mm²/Hz)
450	5.480957	17.69594

原表索引	频率 (Hz)	PSD (mm²/Hz)
451	5.493164	17.78552
452	5.505371	17.876
453	5.517578	17.96738
454	5.529785	18.05967
455	5.541992	18.15289
456	5.554199	18.24705
457	5.566406	18.34216
458	5.578613	18.43824
459	5.59082	18.53529
460	5.603027	18.63333
461	5.615234	18.73238
462	5.627441	18.83245
463	5.639648	18.93355
464	5.651855	19.03569
465	5.664063	19.1389
466	5.67627	19.24318
467	5.688477	19.34855
468	5.700684	19.45503
469	5.712891	19.56262
470	5.725098	19.67136
471	5.737305	19.78125
472	5.749512	19.89231
473	5.761719	20.00455
474	5.773926	20.118
475	5.786133	20.23266
476	5.79834	20.34857
477	5.810547	20.46573
478	5.822754	20.58417
479	5.834961	20.7039
480	5.847168	20.82494

原表索引	频率 (Hz)	PSD (mm²/Hz)
481	5.859375	20.94731
482	5.871582	21.07103
483	5.883789	21.19612
484	5.895996	21.32261
485	5.908203	21.4505
486	5.92041	21.57983
487	5.932617	21.71061
488	5.944824	21.84287
489	5.957031	21.97663
490	5.969238	22.11191
491	5.981445	22.24873
492	5.993652	22.38713
493	6.005859	22.52712
494	6.018066	22.66872
495	6.030273	22.81197
496	6.04248	22.95689
497	6.054688	23.1035
498	6.066895	23.25183
499	6.079102	23.40191
500	6.091309	23.55377
501	6.103516	23.70743
502	6.115723	23.86293
503	6.12793	24.02029
504	6.140137	24.17954
505	6.152344	24.34072
506	6.164551	24.50385
507	6.176758	24.66897
508	6.188965	24.83611
509	6.201172	25.00531
510	6.213379	25.17659
511	6.225586	25.34999
512	6.237793	25.52556

原表索引	频率 (Hz)	PSD (mm²/Hz)
513	6.25	25.70331
514	6.262207	25.8833
515	6.274414	26.06556
516	6.286621	26.25012
517	6.298828	26.43704
518	6.311035	26.62634
519	6.323242	26.81806
520	6.335449	27.01226
521	6.347656	27.20898
522	6.359863	27.40825
523	6.37207	27.61012
524	6.384277	27.81465
525	6.396484	28.02187
526	6.408691	28.23183
527	6.420898	28.44459
528	6.433105	28.66019
529	6.445313	28.87869
530	6.45752	29.10013
531	6.469727	29.32457
532	6.481934	29.55208
533	6.494141	29.78269
534	6.506348	30.01648
535	6.518555	30.25349
536	6.530762	30.4938
537	6.542969	30.73745
538	6.555176	30.98453
539	6.567383	31.23508
540	6.57959	31.48918
541	6.591797	31.74689
542	6.604004	32.00829
543	6.616211	32.27344
544	6.628418	32.54242

原表索引	频率 (Hz)	PSD (mm²/Hz)
545	6.640625	32.81531
546	6.652832	33.09218
547	6.665039	33.3731
548	6.677246	33.65817
549	6.689453	33.94745
550	6.70166	34.24105
551	6.713867	34.53904
552	6.726074	34.84151
553	6.738281	35.14856
554	6.750488	35.46028
555	6.762695	35.77676
556	6.774902	36.0981
557	6.787109	36.42441
558	6.799316	36.75579
559	6.811523	37.09234
560	6.82373	37.43418
561	6.835938	37.78142
562	6.848145	38.13418
563	6.860352	38.49256
564	6.872559	38.8567
565	6.884766	39.22672
566	6.896973	39.60275
567	6.90918	39.98491
568	6.921387	40.37336
569	6.933594	40.76822
570	6.945801	41.16964
571	6.958008	41.57777
572	6.970215	41.99277
573	6.982422	42.41477
574	6.994629	42.84396
575	7.006836	43.28049
576	7.019043	43.72454

原表索引	频率 (Hz)	PSD (mm²/Hz)
577	7.03125	44.17627
578	7.043457	44.63587
579	7.055664	45.10353
580	7.067871	45.57944
581	7.080078	46.06379
582	7.092285	46.55679
583	7.104492	47.05865
584	7.116699	47.56959
585	7.128906	48.08981
586	7.141113	48.61957
587	7.15332	49.15908
588	7.165527	49.70861
589	7.177734	50.26839
590	7.189941	50.83869
591	7.202148	51.41977
592	7.214355	52.01192
593	7.226563	52.61541
594	7.23877	53.23055
595	7.250977	53.85764
596	7.263184	54.49698
597	7.275391	55.14892
598	7.287598	55.81378
599	7.299805	56.49191
600	7.312012	57.18367

原表索引	频率 (Hz)	PSD (mm²/Hz)
601	7.324219	57.88943
602	7.336426	58.60958
603	7.348633	59.34451
604	7.36084	60.09464
605	7.373047	60.86039
606	7.385254	61.6422
607	7.397461	62.44052

原表索引	频率 (Hz)	PSD (mm²/Hz)
608	7.409668	63.25584
609	7.421875	64.08864
610	7.434082	64.93944
611	7.446289	65.80874
612	7.458496	66.69711
613	7.470703	67.60511
614	7.48291	68.53333
615	7.495117	69.48237
616	7.507324	70.45287
617	7.519531	71.44549
618	7.530867	72.3877
619	7.542204	73.35015
620	7.55354	74.3334
621	7.564876	75.33809
622	7.576212	76.36484
623	7.587548	77.4143
624	7.598885	78.48715
625	7.610221	79.5841
626	7.611092	79.66939
627	7.611963	79.75482
628	7.612833	79.8404
629	7.613704	79.92612
630	7.614575	80.01199
631	7.615446	80.09801
632	7.616317	80.18418
633	7.617188	80.2705
634	7.629395	81.49637
635	7.641602	82.75254
636	7.653809	84.04004
637	7.666016	85.35991
638	7.678223	86.71326
639	7.69043	88.10124

原表索引	频率 (Hz)	PSD (mm²/Hz)
640	7.702637	89.52505
641	7.714844	90.98593
642	7.727051	92.4852
643	7.739258	94.02421
644	7.751465	95.6044
645	7.763672	97.22725
646	7.775879	98.89432
647	7.788086	100.6072
648	7.800293	102.3677
649	7.8125	104.1775
650	7.824707	106.0385
651	7.836914	107.9527
652	7.849121	109.9221
653	7.861328	111.9489
654	7.873535	114.0352
655	7.885742	116.1836
656	7.897949	118.3964
657	7.910156	120.6764
658	7.922363	123.0261
659	7.93457	125.4486
660	7.946777	127.9469
661	7.958984	130.5241
662	7.971191	133.1836
663	7.983398	135.9289
664	7.995605	138.7639
665	8.007813	141.6924
666	8.02002	144.7186
667	8.032227	147.8469
668	8.044434	151.0819
669	8.056641	154.4285
670	8.068848	157.892
671	8.081055	161.4777

原表索引	频率 (Hz)	PSD (mm²/Hz)
672	8.093262	165.1916
673	8.105469	169.0397
674	8.117676	173.0286
675	8.129883	177.1653
676	8.14209	181.457
677	8.154297	185.9116
678	8.166504	190.5373
679	8.178711	195.3431
680	8.190918	200.3382
681	8.203125	205.5328
682	8.215332	210.9375
683	8.227539	216.5636
684	8.239746	222.4234
685	8.251953	228.5298
686	8.26416	234.8968
687	8.276367	241.5391
688	8.288574	248.4727
689	8.300781	255.7146
690	8.305059	258.3288
691	8.309337	260.984
692	8.313615	263.6809
693	8.317893	266.4204
694	8.32217	269.2035
695	8.326448	272.031
696	8.330726	274.904
697	8.335004	277.8233
698	8.342933	283.3603
699	8.350862	289.0665
700	8.358791	294.9488
701	8.366721	301.0142
702	8.37465	307.2704
703	8.382579	313.7253

原表索引	频率 (Hz)	PSD (mm²/Hz)
704	8.390508	320.3872
705	8.398438	327.2649
706	8.410645	338.2963
707	8.422852	349.8971
708	8.435059	362.106
709	8.447266	374.9652
710	8.459473	388.5205
711	8.47168	402.8216
712	8.483887	417.9223
713	8.496094	433.8816
714	8.508301	450.7633
715	8.520508	468.6373
716	8.532715	487.5799
717	8.544922	507.6742
718	8.557129	529.0113
719	8.569336	551.6908
720	8.581543	575.8219
721	8.59375	601.5244
722	8.605957	628.9296
723	8.618164	658.1817
724	8.630371	689.439
725	8.642578	722.8757
726	8.654785	758.6831
727	8.666992	797.0715
728	8.679199	838.2719
729	8.691406	882.5382
730	8.692155	885.3588
731	8.692904	888.1921
732	8.693652	891.0381
733	8.694401	893.8968
734	8.695149	896.7684
735	8.695898	899.6529

原表索引	频率 (Hz)	PSD (mm²/Hz)
736	8.696647	902.5504
737	8.697395	905.4609
738	8.708854	951.6886
739	8.720312	1001.254
740	8.731771	1054.444
741	8.743229	1111.566
742	8.754687	1172.956
743	8.766146	1238.97
744	8.777604	1309.991
745	8.789063	1386.425
746	8.800254	1466.705
747	8.811445	1552.959
748	8.822636	1645.597
749	8.833827	1745.026
750	8.845018	1851.629

原表索引	频率 (Hz)	PSD (mm²/Hz)
751	8.856209	1965.749
752	8.8674	2087.659
753	8.878591	2217.529
754	8.879607	2229.715
755	8.880623	2241.967
756	8.881639	2254.285
757	8.882655	2266.669
758	8.883671	2279.118
759	8.884687	2291.633
760	8.885703	2304.213
761	8.886719	2316.858
762	8.898926	2473.815
763	8.911133	2639.635
764	8.92334	2813.357
765	8.935547	2993.437
766	8.947754	3177.626

原表索引	频率 (Hz)	PSD (mm²/Hz)
767	8.959961	3362.868
768	8.972168	3545.243
769	8.984375	3719.982
770	8.993801	3846.245
771	9.003228	3961.964
772	9.012654	4064.354
773	9.022081	4150.71
774	9.031507	4218.561
775	9.040934	4265.829
776	9.05036	4290.978
777	9.059787	4293.13
778	9.062567	4289.332
779	9.065348	4283.523
780	9.068128	4275.718
781	9.070909	4265.937
782	9.07369	4254.205
783	9.07647	4240.553
784	9.079251	4225.015
785	9.082031	4207.633
786	9.094238	4110.713
787	9.106445	3983.861
788	9.118652	3832.727
789	9.130859	3663.396
790	9.143066	3481.869
791	9.155273	3293.65
792	9.16748	3103.491
793	9.179688	2915.269
794	9.187349	2799.492
795	9.195011	2686.244
796	9.202673	2575.979
797	9.210335	2469.045
798	9.217997	2365.698

原表索引	频率 (Hz)	PSD (mm²/Hz)
799	9.225659	2266.112
800	9.233321	2170.393
801	9.240983	2078.591
802	9.245528	2025.985
803	9.250073	1974.751
804	9.254618	1924.88
805	9.259163	1876.357
806	9.263708	1829.164
807	9.268253	1783.282
808	9.272799	1738.689
809	9.277344	1695.361
810	9.289551	1585.051
811	9.301758	1483.145
812	9.313965	1389.081
813	9.326172	1302.292
814	9.338379	1222.222
815	9.350586	1148.335
816	9.362793	1080.125
817	9.375	1017.118
818	9.380897	988.4106
819	9.386795	960.7677
820	9.392692	934.1441
821	9.398589	908.4969
822	9.404486	883.7847
823	9.410384	859.968
824	9.416281	837.009
825	9.422178	814.8713
826	9.428488	792.0557
827	9.434798	770.1003
828	9.441108	748.9666
829	9.447417	728.6178
830	9.453727	709.0191

原表索引	频率 (Hz)	PSD (mm²/Hz)
831	9.460037	690.1372
832	9.466347	671.9404
833	9.472656	654.3987
834	9.484863	622.216
835	9.49707	592.1862
836	9.509277	564.1341
837	9.521484	537.9005
838	9.533691	513.341
839	9.545898	490.3239
840	9.558105	468.7297
841	9.570313	448.4494
842	9.58252	429.3836
843	9.594727	411.4418
844	9.606934	394.5413
845	9.619141	378.6065
846	9.631348	363.5681
847	9.643555	349.3629
848	9.655762	335.9327
849	9.667969	323.2243
850	9.680176	311.1887
851	9.692383	299.7808
852	9.70459	288.9593
853	9.716797	278.6859
854	9.729004	268.9254
855	9.741211	259.6453
856	9.753418	250.8155
857	9.765625	242.4082
858	9.767993	240.824
859	9.770361	239.2545
860	9.772729	237.6995
861	9.775097	236.159
862	9.777465	234.6327

原表索引	频率 (Hz)	PSD (mm²/Hz)
863	9.779834	233.1204
864	9.782202	231.622
865	9.78457	230.1373
866	9.794409	224.1123
867	9.804248	218.3101
868	9.814087	212.7204
869	9.823925	207.3329
870	9.833764	202.1385
871	9.843603	197.1282
872	9.853442	192.2937
873	9.863281	187.627
874	9.875488	182.0591
875	9.887695	176.7246
876	9.899902	171.611
877	9.912109	166.7065
878	9.924316	162.0001
879	9.936523	157.4816
880	9.94873	153.1412
881	9.960938	148.9701
882	9.973145	144.9596
883	9.985352	141.1018
884	9.997559	137.3892
885	10.00977	133.8149
886	10.02197	130.3721
887	10.03418	127.0546
888	10.04639	123.8565
889	10.05859	120.7723
890	10.0708	117.7968
891	10.08301	114.925
892	10.09521	112.1523
893	10.10742	109.4741
894	10.11963	106.8865

原表索引	频率 (Hz)	PSD (mm²/Hz)
895	10.13184	104.3854
896	10.14404	101.967
897	10.15625	99.62795
898	10.16846	97.36475
899	10.18066	95.17425
900	10.19287	93.05344

原表索引	频率 (Hz)	PSD (mm²/Hz)
901	10.20508	90.99944
902	10.21729	89.00954
903	10.22949	87.08115
904	10.2417	85.21182
905	10.25391	83.39919
906	10.26611	81.64105
907	10.27832	79.93527
908	10.29053	78.27984
909	10.30273	76.67283
910	10.31494	75.11242
911	10.32715	73.59684
912	10.33936	72.12444
913	10.35156	70.69362
914	10.36377	69.30286
915	10.37598	67.95071
916	10.38818	66.63577
917	10.40039	65.35673
918	10.4126	64.11232
919	10.4248	62.90131
920	10.43701	61.72255
921	10.44922	60.57493
922	10.45674	59.88329
923	10.46425	59.2028
924	10.47177	58.53324
925	10.47929	57.87438

原表索引	频率 (Hz)	PSD (mm²/Hz)
926	10.4868	57.22599
927	10.49432	56.58787
928	10.50184	55.95979
929	10.50935	55.34156
930	10.51404	54.96069
931	10.51873	54.58354
932	10.52342	54.21004
933	10.52811	53.84015
934	10.5328	53.47383
935	10.53749	53.11104
936	10.54218	52.75172
937	10.54688	52.39585
938	10.55908	51.48543
939	10.57129	50.59726
940	10.5835	49.73064
941	10.5957	48.88489
942	10.60791	48.05936
943	10.62012	47.25342
944	10.63232	46.46647
945	10.64453	45.69793
946	10.65674	44.94724
947	10.66895	44.21386
948	10.68115	43.49728
949	10.69336	42.797
950	10.70557	42.11254
951	10.71777	41.44343
952	10.72998	40.78924
953	10.74219	40.14952
954	10.75439	39.52386
955	10.7666	38.91187
956	10.77881	38.31315
957	10.79102	37.72734

原表索引	频率 (Hz)	PSD (mm²/Hz)
958	10.80322	37.15407
959	10.81543	36.593
960	10.82764	36.04378
961	10.83984	35.5061
962	10.85205	34.97964
963	10.86426	34.4641
964	10.87646	33.95918
965	10.88867	33.4646
966	10.90088	32.98009
967	10.91309	32.50538
968	10.92529	32.04022
969	10.9375	31.58435
970	10.94971	31.13754
971	10.96191	30.69955
972	10.97412	30.27015
973	10.98633	29.84914
974	10.99854	29.43629
975	11.01074	29.0314
976	11.02295	28.63428
977	11.03516	28.24473
978	11.04736	27.86256
979	11.05957	27.4876
980	11.07178	27.11966
981	11.08398	26.75858
982	11.09619	26.40419
983	11.1084	26.05633
984	11.12061	25.71485
985	11.13281	25.37959
986	11.14502	25.0504
987	11.15723	24.72716
988	11.16943	24.40971
989	11.18164	24.09792

原表索引	频率 (Hz)	PSD (mm²/Hz)
990	11.19385	23.79166
991	11.20605	23.49081
992	11.21826	23.19524
993	11.23047	22.90483
994	11.24268	22.61946
995	11.25488	22.33903
996	11.26709	22.06342
997	11.2793	21.79252
998	11.2915	21.52623
999	11.30371	21.26446
1000	11.31592	21.00709
1001	11.32813	20.75403
1002	11.34033	20.5052
1003	11.35254	20.2605
1004	11.36475	20.01983
1005	11.37695	19.78313
1006	11.38916	19.55029
1007	11.40137	19.32124
1008	11.41357	19.0959
1009	11.42578	18.8742
1010	11.43799	18.65605
1011	11.4502	18.44138
1012	11.4624	18.23013
1013	11.47461	18.02221
1014	11.48682	17.81757
1015	11.49902	17.61614
1016	11.51123	17.41784
1017	11.52344	17.22263
1018	11.53564	17.03043
1019	11.54785	16.84119
1020	11.56006	16.65485
1021	11.57227	16.47134

原表索引	频率 (Hz)	PSD (mm²/Hz)
1022	11.58447	16.29062
1023	11.59668	16.11264
1024	11.60889	15.93733
1025	11.62109	15.76464
1026	11.6333	15.59453
1027	11.64551	15.42694
1028	11.65771	15.26183
1029	11.66992	15.09915
1030	11.68213	14.93885
1031	11.69434	14.7809
1032	11.70654	14.62523
1033	11.71875	14.47182
1034	11.73096	14.32062
1035	11.74316	14.17158
1036	11.75537	14.02468
1037	11.76758	13.87986
1038	11.77979	13.73709
1039	11.79199	13.59634
1040	11.8042	13.45757
1041	11.81641	13.32073
1042	11.82861	13.18581
1043	11.84082	13.05275
1044	11.85303	12.92154
1045	11.86523	12.79213
1046	11.87744	12.66449
1047	11.88965	12.5386
1048	11.90186	12.41441
1049	11.91406	12.29192
1050	11.92627	12.17107

原表索引	频率 (Hz)	PSD (mm²/Hz)
1051	11.93848	12.05185
1052	11.95068	11.93422

原表索引	频率 (Hz)	PSD (mm²/Hz)
1053	11.96289	11.81816
1054	11.9751	11.70365
1055	11.9873	11.59065
1056	11.99951	11.47914
1057	12.01172	11.36909
1058	12.02393	11.26048
1059	12.03613	11.15329
1060	12.04834	11.04749
1061	12.06055	10.94306
1062	12.07275	10.83997
1063	12.08496	10.73821
1064	12.09717	10.63775
1065	12.10938	10.53857
1066	12.12158	10.44065
1067	12.13379	10.34397
1068	12.146	10.2485
1069	12.1582	10.15424
1070	12.17041	10.06115
1071	12.18262	9.969227
1072	12.19482	9.878444
1073	12.20703	9.788785
1074	12.21924	9.700232
1075	12.23145	9.612767
1076	12.24365	9.526374
1077	12.25586	9.441035
1078	12.26807	9.356734
1079	12.28027	9.273455
1080	12.29248	9.191182
1081	12.30469	9.109898
1082	12.31689	9.02959
1083	12.3291	8.950242
1084	12.34131	8.871839

原表索引	频率 (Hz)	PSD (mm²/Hz)
1085	12.35352	8.794367
1086	12.36572	8.717812
1087	12.37793	8.642159
1088	12.39014	8.567396
1089	12.40234	8.493508
1090	12.41455	8.420483
1091	12.42676	8.348307
1092	12.43896	8.276969
1093	12.45117	8.206455
1094	12.46338	8.136753
1095	12.47559	8.067852
1096	12.48779	7.99974
1097	12.5	7.932404
1098	12.51221	7.865834
1099	12.52441	7.800019
1100	12.53662	7.734947
1101	12.54883	7.670609
1102	12.56104	7.606992
1103	12.57324	7.544087
1104	12.58545	7.481884
1105	12.59766	7.420373
1106	12.60986	7.359544
1107	12.62207	7.299387
1108	12.63428	7.239892
1109	12.64648	7.181051
1110	12.65869	7.122854
1111	12.6709	7.065292
1112	12.68311	7.008357
1113	12.69531	6.952039
1114	12.70752	6.89633
1115	12.71973	6.841221
1116	12.73193	6.786705

原表索引	频率 (Hz)	PSD (mm²/Hz)
1117	12.74414	6.732773
1118	12.75635	6.679418
1119	12.76855	6.62663
1120	12.78076	6.574403
1121	12.79297	6.52273
1122	12.80518	6.471601
1123	12.81738	6.421011
1124	12.82959	6.370952
1125	12.8418	6.321416
1126	12.854	6.272398
1127	12.86621	6.223889
1128	12.87842	6.175883
1129	12.89063	6.128374
1130	12.90283	6.081355
1131	12.91504	6.034819
1132	12.92725	5.988761
1133	12.93945	5.943173
1134	12.95166	5.89805
1135	12.96387	5.853386
1136	12.97607	5.809174
1137	12.98828	5.76541
1138	13.00049	5.722086
1139	13.0127	5.679198
1140	13.0249	5.63674
1141	13.03711	5.594707
1142	13.04932	5.553092
1143	13.06152	5.511891
1144	13.07373	5.471098
1145	13.08594	5.430709
1146	13.09814	5.390718
1147	13.11035	5.35112
1148	13.12256	5.31191

原表索引	频率 (Hz)	PSD (mm²/Hz)
1149	13.13477	5.273084
1150	13.14697	5.234637
1151	13.15918	5.196563
1152	13.17139	5.158859
1153	13.18359	5.121519
1154	13.1958	5.08454
1155	13.20801	5.047916
1156	13.22021	5.011644
1157	13.23242	4.975719
1158	13.24463	4.940137
1159	13.25684	4.904894
1160	13.26904	4.869985
1161	13.28125	4.835408
1162	13.29346	4.801156
1163	13.30566	4.767228
1164	13.31787	4.733618
1165	13.33008	4.700323
1166	13.34229	4.66734
1167	13.35449	4.634664
1168	13.3667	4.602293
1169	13.37891	4.570221
1170	13.39111	4.538447
1171	13.40332	4.506965
1172	13.41553	4.475774
1173	13.42773	4.444869
1174	13.43994	4.414247
1175	13.45215	4.383905
1176	13.46436	4.353839
1177	13.47656	4.324047
1178	13.48877	4.294526
1179	13.50098	4.265271
1180	13.51318	4.23628

原表索引	频率 (Hz)	PSD (mm²/Hz)
1181	13.52539	4.207551
1182	13.5376	4.179079
1183	13.5498	4.150863
1184	13.56201	4.122898
1185	13.57422	4.095183
1186	13.58643	4.067715
1187	13.59863	4.04049
1188	13.61084	4.013507
1189	13.62305	3.986761
1190	13.63525	3.960251
1191	13.64746	3.933975
1192	13.65967	3.907928
1193	13.67188	3.882109
1194	13.68408	3.856516
1195	13.69629	3.831146
1196	13.7085	3.805995
1197	13.7207	3.781063
1198	13.73291	3.756346
1199	13.74512	3.731843
1200	13.75732	3.70755

原表索引	频率 (Hz)	PSD (mm²/Hz)
1201	13.76953	3.683466
1202	13.78174	3.659588
1203	13.79395	3.635914
1204	13.80615	3.612442
1205	13.81836	3.589169
1206	13.83057	3.566094
1207	13.84277	3.543215
1208	13.85498	3.520528
1209	13.86719	3.498033
1210	13.87939	3.475727
1211	13.8916	3.453608

原表索引	频率 (Hz)	PSD (mm²/Hz)
1212	13.90381	3.431675
1213	13.91602	3.409924
1214	13.92822	3.388355
1215	13.94043	3.366965
1216	13.95264	3.345753
1217	13.96484	3.324716
1218	13.97705	3.303853
1219	13.98926	3.283162
1220	14.00146	3.262641
1221	14.01367	3.242289
1222	14.02588	3.222103
1223	14.03809	3.202082
1224	14.05029	3.182224
1225	14.0625	3.162528
1226	14.07471	3.142991
1227	14.08691	3.123613
1228	14.09912	3.104392
1229	14.11133	3.085325
1230	14.12354	3.066412
1231	14.13574	3.047651
1232	14.14795	3.02904
1233	14.16016	3.010578
1234	14.17236	2.992263
1235	14.18457	2.974094
1236	14.19678	2.956069
1237	14.20898	2.938188
1238	14.22119	2.920448
1239	14.2334	2.902847
1240	14.24561	2.885386
1241	14.25781	2.868062
1242	14.27002	2.850873
1243	14.28223	2.83382

原表索引	频率 (Hz)	PSD (mm²/Hz)
1244	14.29443	2.816899
1245	14.30664	2.800111
1246	14.31885	2.783453
1247	14.33105	2.766924
1248	14.34326	2.750524
1249	14.35547	2.73425
1250	14.36768	2.718102
1251	14.37988	2.702079
1252	14.39209	2.686178
1253	14.4043	2.6704
1254	14.4165	2.654742
1255	14.42871	2.639204
1256	14.44092	2.623785
1257	14.45313	2.608482
1258	14.46533	2.593296
1259	14.47754	2.578225
1260	14.48975	2.563269
1261	14.50195	2.548425
1262	14.51416	2.533693
1263	14.52637	2.519071
1264	14.53857	2.50456
1265	14.55078	2.490157
1266	14.56299	2.475861
1267	14.5752	2.461673
1268	14.5874	2.44759
1269	14.59961	2.433612
1270	14.61182	2.419737
1271	14.62402	2.405965
1272	14.63623	2.392295
1273	14.64844	2.378726
1274	14.66064	2.365257
1275	14.67285	2.351886

原表索引	频率 (Hz)	PSD (mm²/Hz)
1276	14.68506	2.338614
1277	14.69727	2.325438
1278	14.70947	2.312359
1279	14.72168	2.299376
1280	14.73389	2.286487
1281	14.74609	2.273691
1282	14.7583	2.260988
1283	14.77051	2.248377
1284	14.78271	2.235858
1285	14.79492	2.223428
1286	14.80713	2.211088
1287	14.81934	2.198837
1288	14.83154	2.186673
1289	14.84375	2.174597
1290	14.85596	2.162606
1291	14.86816	2.150701
1292	14.88037	2.138881
1293	14.89258	2.127145
1294	14.90479	2.115492
1295	14.91699	2.103922
1296	14.9292	2.092433
1297	14.94141	2.081025
1298	14.95361	2.069698
1299	14.96582	2.05845
1300	14.97803	2.047281
1301	14.99023	2.03619
1302	15.00244	2.025177
1303	15.01465	2.014241
1304	15.02686	2.00338
1305	15.03906	1.992596
1306	15.05127	1.981886
1307	15.06348	1.97125

原表索引	频率 (Hz)	PSD (mm²/Hz)
1308	15.07568	1.960688
1309	15.08789	1.950198
1310	15.1001	1.939781
1311	15.1123	1.929436
1312	15.12451	1.919162
1313	15.13672	1.908958
1314	15.14718	1.900265
1315	15.15765	1.891623
1316	15.16811	1.883032
1317	15.17858	1.874492
1318	15.18905	1.866
1319	15.19951	1.857559
1320	15.20998	1.849166
1321	15.22044	1.840823
1322	15.22218	1.839439
1323	15.22393	1.838056
1324	15.22567	1.836675
1325	15.22741	1.835295
1326	15.22915	1.833917
1327	15.23089	1.83254
1328	15.23263	1.831164
1329	15.23438	1.829789
1330	15.24658	1.820192
1331	15.25879	1.81066
1332	15.271	1.801192
1333	15.2832	1.791788
1334	15.29541	1.782446
1335	15.30762	1.773168
1336	15.31982	1.763951
1337	15.33203	1.754795
1338	15.34424	1.7457
1339	15.35645	1.736666

原表索引	频率 (Hz)	PSD (mm²/Hz)
1340	15.36865	1.727692
1341	15.38086	1.718777
1342	15.39307	1.709921
1343	15.40527	1.701124
1344	15.41748	1.692385
1345	15.42969	1.683703
1346	15.44189	1.675078
1347	15.4541	1.666509
1348	15.46631	1.657997
1349	15.47852	1.64954
1350	15.49072	1.641139

原表索引	频率 (Hz)	PSD (mm²/Hz)
1351	15.50293	1.632792
1352	15.51514	1.6245
1353	15.52734	1.616261
1354	15.53955	1.608076
1355	15.55176	1.599943
1356	15.56396	1.591864
1357	15.57617	1.583836
1358	15.58838	1.57586
1359	15.60059	1.567935
1360	15.61279	1.560062
1361	15.625	1.552238
1362	15.63721	1.544465
1363	15.64941	1.536741
1364	15.66162	1.529067
1365	15.67383	1.521442
1366	15.68604	1.513865
1367	15.69824	1.506336
1368	15.71045	1.498855
1369	15.72266	1.491421
1370	15.73486	1.484034

原表索引	频率 (Hz)	PSD (mm²/Hz)
1371	15.74707	1.476693
1372	15.75928	1.469399
1373	15.77148	1.462151
1374	15.78369	1.454948
1375	15.7959	1.447791
1376	15.80811	1.440678
1377	15.82031	1.43361
1378	15.83252	1.426586
1379	15.84473	1.419605
1380	15.85693	1.412668
1381	15.86914	1.405774
1382	15.88135	1.398923
1383	15.89355	1.392114
1384	15.90576	1.385347
1385	15.91797	1.378622
1386	15.93018	1.371939
1387	15.94238	1.365296
1388	15.95459	1.358695
1389	15.9668	1.352134
1390	15.979	1.345613
1391	15.99121	1.339132
1392	16.00342	1.33269
1393	16.01563	1.326288
1394	16.02783	1.319925
1395	16.04004	1.313601
1396	16.05225	1.307314
1397	16.06445	1.301066
1398	16.07666	1.294856
1399	16.08887	1.288684
1400	16.10107	1.282548
1401	16.11328	1.27645
1402	16.12549	1.270388

原表索引	频率 (Hz)	PSD (mm²/Hz)
1403	16.1377	1.264363
1404	16.1499	1.258373
1405	16.16211	1.25242
1406	16.17432	1.246502
1407	16.18652	1.24062
1408	16.19873	1.234773
1409	16.21094	1.22896
1410	16.22314	1.223182
1411	16.23535	1.217439
1412	16.24756	1.211729
1413	16.25977	1.206053
1414	16.27197	1.200411
1415	16.28418	1.194802
1416	16.29639	1.189226
1417	16.30859	1.183683
1418	16.3208	1.178173
1419	16.33301	1.172695
1420	16.34521	1.167249
1421	16.35742	1.161835
1422	16.36963	1.156452
1423	16.38184	1.151101
1424	16.39404	1.145781
1425	16.40625	1.140492
1426	16.41846	1.135234
1427	16.43066	1.130006
1428	16.44287	1.124809
1429	16.45508	1.119642
1430	16.46729	1.114504
1431	16.47949	1.109396
1432	16.4917	1.104318
1433	16.50391	1.099269
1434	16.51611	1.094249

原表索引	频率 (Hz)	PSD (mm²/Hz)
1435	16.52832	1.089257
1436	16.54053	1.084295
1437	16.55273	1.07936
1438	16.56494	1.074454
1439	16.57715	1.069576
1440	16.58936	1.064726
1441	16.60156	1.059903
1442	16.61012	1.056539
1443	16.61867	1.053189
1444	16.62723	1.049852
1445	16.63579	1.046528
1446	16.64434	1.043217
1447	16.6529	1.03992
1448	16.66145	1.036635
1449	16.67001	1.033364
1450	16.67366	1.031972
1451	16.67731	1.030582
1452	16.68096	1.029194
1453	16.68461	1.027809
1454	16.68826	1.026426
1455	16.69192	1.025046
1456	16.69557	1.023668
1457	16.69922	1.022292
1458	16.71143	1.017709
1459	16.72363	1.013152
1460	16.73584	1.00862
1461	16.74805	1.004114
1462	16.76025	0.9996328
1463	16.77246	0.9951768
1464	16.78467	0.9907456
1465	16.79688	0.9863392
1466	16.80908	0.9819572

原表索引	频率 (Hz)	PSD (mm²/Hz)
1467	16.82129	0.9775995
1468	16.8335	0.9732661
1469	16.8457	0.9689566
1470	16.85791	0.964671
1471	16.87012	0.9604091
1472	16.88232	0.9561706
1473	16.89453	0.9519556
1474	16.90674	0.9477637
1475	16.91895	0.9435949
1476	16.93115	0.9394489
1477	16.94336	0.9353257
1478	16.95557	0.931225
1479	16.96777	0.9271468
1480	16.97998	0.9230908
1481	16.99219	0.919057
1482	17.00439	0.9150451
1483	17.0166	0.911055
1484	17.02881	0.9070866
1485	17.04102	0.9031398
1486	17.05322	0.8992143
1487	17.06543	0.8953101
1488	17.07764	0.8914269
1489	17.08984	0.8875647
1490	17.10205	0.8837234
1491	17.11426	0.8799027
1492	17.12646	0.8761026
1493	17.13867	0.8723228
1494	17.15088	0.8685634
1495	17.16309	0.8648241
1496	17.17529	0.8611047
1497	17.1875	0.8574053
1498	17.19971	0.8537256

原表索引	频率 (Hz)	PSD (mm²/Hz)
1499	17.21191	0.8500655
1500	17.22412	0.8464249

原表索引	频率 (Hz)	PSD (mm²/Hz)
1501	17.23633	0.8428036
1502	17.24854	0.8392016
1503	17.26074	0.8356186
1504	17.27295	0.8320547
1505	17.28516	0.8285096
1506	17.29736	0.8249832
1507	17.30957	0.8214754
1508	17.32178	0.8179861
1509	17.33398	0.8145152
1510	17.34619	0.8110625
1511	17.3584	0.807628
1512	17.37061	0.8042114
1513	17.38281	0.8008128
1514	17.38431	0.8003971
1515	17.38581	0.7999818
1516	17.3873	0.7995667
1517	17.3888	0.7991518
1518	17.3903	0.7987372
1519	17.3918	0.7983229
1520	17.39329	0.7979089
1521	17.39479	0.7974951
1522	17.4055	0.7945432
1523	17.41621	0.7916048
1524	17.42692	0.7886799
1525	17.43763	0.7857683
1526	17.44834	0.78287
1527	17.45905	0.7799849
1528	17.46976	0.777113
1529	17.48047	0.7742541

原表索引	频率 (Hz)	PSD (mm²/Hz)
1530	17.49268	0.7710114
1531	17.50488	0.7677855
1532	17.51709	0.7645763
1533	17.5293	0.7613836
1534	17.5415	0.7582074
1535	17.55371	0.7550475
1536	17.56592	0.7519039
1537	17.57813	0.7487764
1538	17.59033	0.745665
1539	17.60254	0.7425695
1540	17.61475	0.7394899
1541	17.62695	0.736426
1542	17.63916	0.7333778
1543	17.65137	0.7303451
1544	17.66357	0.7273279
1545	17.67578	0.724326
1546	17.68596	0.7218355
1547	17.69613	0.7193555
1548	17.70631	0.716886
1549	17.71648	0.7144269
1550	17.72666	0.7119782
1551	17.73683	0.7095398
1552	17.74701	0.7071118
1553	17.75718	0.7046939
1554	17.75921	0.7042123
1555	17.76125	0.7037311
1556	17.76328	0.7032503
1557	17.76531	0.7027699
1558	17.76734	0.7022899
1559	17.76937	0.7018103
1560	17.77141	0.7013311
1561	17.77344	0.7008523

原表索引	频率 (Hz)	PSD (mm²/Hz)
1562	17.78564	0.6979843
1563	17.79785	0.6951308
1564	17.81006	0.6922916
1565	17.82227	0.6894667
1566	17.83447	0.6866559
1567	17.84668	0.6838592
1568	17.85889	0.6810765
1569	17.87109	0.6783077
1570	17.8833	0.6755527
1571	17.89551	0.6728114
1572	17.90771	0.6700837
1573	17.91992	0.6673697
1574	17.93213	0.6646691
1575	17.94434	0.6619819
1576	17.95654	0.659308
1577	17.96875	0.6566473
1578	17.98096	0.6539998
1579	17.99316	0.6513654
1580	18.00537	0.648744
1581	18.01758	0.6461355
1582	18.02979	0.6435398
1583	18.04199	0.6409569
1584	18.0542	0.6383867
1585	18.06641	0.635829
1586	18.07305	0.6344419
1587	18.0797	0.6330584
1588	18.08634	0.6316786
1589	18.09299	0.6303025
1590	18.09964	0.62893
1591	18.10628	0.6275612
1592	18.11293	0.6261961
1593	18.11957	0.6248346

原表索引	频率 (Hz)	PSD (mm²/Hz)
1594	18.12513	0.6236981
1595	18.1307	0.6225641
1596	18.13626	0.6214326
1597	18.14182	0.6203037
1598	18.14738	0.6191773
1599	18.15294	0.6180533
1600	18.1585	0.6169319
1601	18.16406	0.6158129
1602	18.17627	0.6133654
1603	18.18848	0.6109298
1604	18.20068	0.6085059
1605	18.21289	0.6060938
1606	18.2251	0.6036934
1607	18.2373	0.6013046
1608	18.24951	0.5989272
1609	18.26172	0.5965614
1610	18.27393	0.5942069
1611	18.28613	0.5918637
1612	18.29834	0.5895318
1613	18.31055	0.5872111
1614	18.32275	0.5849015
1615	18.33496	0.582603
1616	18.34717	0.5803155
1617	18.35938	0.5780388
1618	18.37158	0.5757731
1619	18.38379	0.5735182
1620	18.396	0.571274
1621	18.4082	0.5690404
1622	18.42041	0.5668175
1623	18.43262	0.5646051
1624	18.44482	0.5624033
1625	18.45703	0.5602118

原表索引	频率 (Hz)	PSD (mm²/Hz)
1626	18.46015	0.559654
1627	18.46326	0.5590968
1628	18.46638	0.5585403
1629	18.4695	0.5579844
1630	18.47261	0.5574293
1631	18.47573	0.5568748
1632	18.47885	0.5563209
1633	18.48197	0.5557678
1634	18.49106	0.5541582
1635	18.50015	0.5525544
1636	18.50924	0.5509561
1637	18.51833	0.5493635
1638	18.52742	0.5477764
1639	18.53651	0.546195
1640	18.5456	0.544619
1641	18.55469	0.5430486
1642	18.56689	0.5409483
1643	18.5791	0.5388579
1644	18.59131	0.5367773
1645	18.60352	0.5347064
1646	18.61572	0.5326452
1647	18.62793	0.5305936
1648	18.64014	0.5285516
1649	18.65234	0.5265191
1650	18.66455	0.524496

原表索引	频率 (Hz)	PSD (mm²/Hz)
1651	18.67676	0.5224824
1652	18.68896	0.5204781
1653	18.70117	0.5184831
1654	18.71338	0.5164973
1655	18.72559	0.5145207
1656	18.73779	0.5125533

原表索引	频率 (Hz)	PSD (mm²/Hz)
1657	18.75	0.510595
1658	18.76179	0.5087114
1659	18.77359	0.5068362
1660	18.78538	0.5049693
1661	18.79718	0.5031108
1662	18.80897	0.5012605
1663	18.82077	0.4994184
1664	18.83256	0.4975845
1665	18.84436	0.4957587
1666	18.84477	0.495695
1667	18.84518	0.4956313
1668	18.84559	0.4955676
1669	18.84601	0.4955039
1670	18.84642	0.4954403
1671	18.84683	0.4953766
1672	18.84724	0.495313
1673	18.84766	0.4952493
1674	18.85986	0.4933705
1675	18.87207	0.4915003
1676	18.88428	0.4896386
1677	18.89648	0.4877854
1678	18.90869	0.4859407
1679	18.9209	0.4841044
1680	18.93311	0.4822764
1681	18.94531	0.4804567
1682	18.95752	0.4786453
1683	18.96973	0.4768421
1684	18.98193	0.4750471
1685	18.99414	0.4732602
1686	19.00635	0.4714814
1687	19.01855	0.4697106
1688	19.03076	0.4679478

原表索引	频率 (Hz)	PSD (mm²/Hz)
1689	19.04297	0.466193
1690	19.05518	0.4644461
1691	19.06738	0.462707
1692	19.07959	0.4609757
1693	19.0918	0.4592523
1694	19.104	0.4575365
1695	19.11621	0.4558285
1696	19.12842	0.4541281
1697	19.14063	0.4524353
1698	19.15283	0.45075
1699	19.16504	0.4490723
1700	19.17725	0.4474021
1701	19.18945	0.4457393
1702	19.20166	0.444084
1703	19.21387	0.4424359
1704	19.22607	0.4407953
1705	19.23828	0.4391618
1706	19.25049	0.4375357
1707	19.2627	0.4359167
1708	19.2749	0.4343049
1709	19.28711	0.4327003
1710	19.29932	0.4311027
1711	19.31152	0.4295122
1712	19.32373	0.4279286
1713	19.33594	0.4263521
1714	19.34814	0.4247825
1715	19.36035	0.4232198
1716	19.37256	0.421664
1717	19.38477	0.4201149
1718	19.39697	0.4185727
1719	19.40918	0.4170373
1720	19.42139	0.4155085

原表索引	频率 (Hz)	PSD (mm²/Hz)
1721	19.43359	0.4139865
1722	19.4458	0.412471
1723	19.45801	0.4109622
1724	19.47021	0.40946
1725	19.48242	0.4079643
1726	19.49463	0.4064752
1727	19.50684	0.4049925
1728	19.51904	0.4035162
1729	19.53125	0.4020463
1730	19.53599	0.4014778
1731	19.54072	0.4009102
1732	19.54546	0.4003435
1733	19.55019	0.3997778
1734	19.55493	0.399213
1735	19.55967	0.3986492
1736	19.5644	0.3980864
1737	19.56914	0.3975245
1738	19.57661	0.39664
1739	19.58408	0.3957579
1740	19.59155	0.3948782
1741	19.59902	0.3940007
1742	19.60649	0.3931256
1743	19.61396	0.3922528
1744	19.62144	0.3913823
1745	19.62891	0.3905141
1746	19.64111	0.3891004
1747	19.65332	0.3876927
1748	19.66553	0.3862912
1749	19.67773	0.3848956
1750	19.68994	0.383506
1751	19.70215	0.3821224
1752	19.71436	0.3807447

原表索引	频率 (Hz)	PSD (mm²/Hz)
1753	19.72656	0.3793728
1754	19.73877	0.3780069
1755	19.75098	0.3766467
1756	19.76318	0.3752924
1757	19.77539	0.3739438
1758	19.7876	0.372601
1759	19.7998	0.3712639
1760	19.81201	0.3699325
1761	19.82422	0.3686067
1762	19.83643	0.3672865
1763	19.84863	0.365972
1764	19.86084	0.364663
1765	19.87305	0.3633595
1766	19.88525	0.3620615
1767	19.89746	0.360769
1768	19.90967	0.359482
1769	19.92188	0.3582004
1770	19.93408	0.3569242
1771	19.94629	0.3556533
1772	19.9585	0.3543878
1773	19.9707	0.3531276
1774	19.98291	0.3518727
1775	19.99512	0.3506231
1776	20.00732	0.3493787
1777	20.01953	0.3481395
1778	20.03174	0.3469055
1779	20.04395	0.3456766
1780	20.05615	0.3444529
1781	20.06836	0.3432342
1782	20.08057	0.3420207
1783	20.09277	0.3408121
1784	20.10498	0.3396087

原表索引	频率 (Hz)	PSD (mm²/Hz)
1785	20.11719	0.3384102
1786	20.12939	0.3372167
1787	20.1416	0.3360281
1788	20.15381	0.3348445
1789	20.16602	0.3336657
1790	20.17822	0.3324919
1791	20.19043	0.3313229
1792	20.20264	0.3301587
1793	20.21484	0.3289993
1794	20.22705	0.3278447
1795	20.23926	0.3266949
1796	20.25146	0.3255498
1797	20.26367	0.3244094
1798	20.27588	0.3232736
1799	20.28809	0.3221426
1800	20.30029	0.3210162

原表索引	频率 (Hz)	PSD (mm²/Hz)
1801	20.3125	0.3198944
1802	20.32471	0.3187772
1803	20.33691	0.3176645
1804	20.34912	0.3165564
1805	20.36133	0.3154529
1806	20.37354	0.3143538
1807	20.38574	0.3132592
1808	20.39795	0.3121691
1809	20.41016	0.3110834
1810	20.42236	0.3100022
1811	20.43457	0.3089253
1812	20.44678	0.3078528
1813	20.45898	0.3067846
1814	20.47119	0.3057208
1815	20.4834	0.3046613
1816	20.49561	0.303606
1817	20.50781	0.3025551
1818	20.52002	0.3015084
1819	20.53223	0.3004659
1820	20.54443	0.2994276
1821	20.55664	0.2983935
1822	20.56885	0.2973635
1823	20.58105	0.2963377
1824	20.59326	0.295316
1825	20.60547	0.2942985
1826	20.61768	0.293285
1827	20.62988	0.2922755
1828	20.64209	0.2912702
1829	20.6543	0.2902688
1830	20.6665	0.2892714
1831	20.67871	0.2882781
1832	20.69092	0.2872887

原表索引	频率 (Hz)	PSD (mm²/Hz)
1833	20.70313	0.2863032
1834	20.71533	0.2853217
1835	20.72754	0.2843441
1836	20.73975	0.2833703
1837	20.75195	0.2824005
1838	20.76416	0.2814345
1839	20.77637	0.2804723
1840	20.78857	0.2795139
1841	20.80078	0.2785594
1842	20.81299	0.2776086
1843	20.8252	0.2766616
1844	20.8374	0.2757183
1845	20.84961	0.2747788
1846	20.86182	0.273843
1847	20.87402	0.2729108
1848	20.88623	0.2719824
1849	20.89844	0.2710575
1850	20.91064	0.2701364
1851	20.92285	0.2692188
1852	20.93506	0.2683049
1853	20.94727	0.2673945
1854	20.95947	0.2664877
1855	20.97168	0.2655845
1856	20.98389	0.2646848
1857	20.99609	0.2637886
1858	20.99892	0.2635816
1859	21.00175	0.2633748
1860	21.00457	0.2631682
1861	21.0074	0.2629618
1862	21.01023	0.2627555
1863	21.01305	0.2625494
1864	21.01588	0.2623436

原表索引	频率 (Hz)	PSD (mm²/Hz)
1865	21.01871	0.2621379
1866	21.02809	0.2614565
1867	21.03747	0.2607773
1868	21.04685	0.2601
1869	21.05623	0.2594248
1870	21.06561	0.2587516
1871	21.07499	0.2580804
1872	21.08437	0.2574112
1873	21.09375	0.256744
1874	21.10596	0.2558788
1875	21.11816	0.255017
1876	21.13037	0.2541584
1877	21.14258	0.2533032
1878	21.15479	0.2524514
1879	21.16699	0.2516028
1880	21.1792	0.2507575
1881	21.19141	0.2499155
1882	21.20361	0.2490767
1883	21.21582	0.2482411
1884	21.22803	0.2474088
1885	21.24023	0.2465797
1886	21.25244	0.2457538
1887	21.26465	0.244931
1888	21.27686	0.2441114
1889	21.28906	0.243295
1890	21.30127	0.2424817
1891	21.31348	0.2416715
1892	21.32568	0.2408644
1893	21.33789	0.2400604
1894	21.3501	0.2392595
1895	21.3623	0.2384616
1896	21.37451	0.2376668

原表索引	频率 (Hz)	PSD (mm²/Hz)
1897	21.38672	0.2368751
1898	21.39893	0.2360863
1899	21.41113	0.2353006
1900	21.42334	0.2345178
1901	21.43555	0.233738
1902	21.44775	0.2329612
1903	21.45996	0.2321874
1904	21.47217	0.2314164
1905	21.48438	0.2306484
1906	21.49658	0.2298834
1907	21.50879	0.2291212
1908	21.521	0.2283619
1909	21.5332	0.2276054
1910	21.54541	0.2268519
1911	21.55762	0.2261012
1912	21.56982	0.2253533
1913	21.58203	0.2246082
1914	21.59424	0.2238659
1915	21.60645	0.2231265
1916	21.61865	0.2223898
1917	21.63086	0.2216559
1918	21.64307	0.2209247
1919	21.65527	0.2201963
1920	21.66748	0.2194706
1921	21.67969	0.2187477
1922	21.69189	0.2180274
1923	21.7041	0.2173099
1924	21.71631	0.216595
1925	21.72852	0.2158828
1926	21.74072	0.2151733
1927	21.75293	0.2144664
1928	21.76514	0.2137622

原表索引	频率 (Hz)	PSD (mm²/Hz)
1929	21.77734	0.2130605
1930	21.78955	0.2123615
1931	21.80176	0.2116651
1932	21.81396	0.2109713
1933	21.82617	0.2102801
1934	21.83838	0.2095914
1935	21.85059	0.2089053
1936	21.86279	0.2082217
1937	21.875	0.2075407
1938	21.88721	0.2068621
1939	21.89941	0.2061861
1940	21.91162	0.2055126
1941	21.92383	0.2048416
1942	21.93604	0.2041731
1943	21.94824	0.203507
1944	21.96045	0.2028434
1945	21.97266	0.2021822
1946	21.98486	0.2015235
1947	21.99707	0.2008671
1948	22.00928	0.2002132
1949	22.02148	0.1995617
1950	22.03369	0.1989126

原表索引	频率 (Hz)	PSD (mm²/Hz)
1951	22.0459	0.1982659
1952	22.05811	0.1976216
1953	22.07031	0.1969796
1954	22.08252	0.1963399
1955	22.09473	0.1957026
1956	22.10693	0.1950676
1957	22.11914	0.194435
1958	22.13135	0.1938047
1959	22.14355	0.1931766

原表索引	频率 (Hz)	PSD (mm²/Hz)
1960	22.15576	0.1925509
1961	22.16797	0.1919274
1962	22.18018	0.1913062
1963	22.19238	0.1906872
1964	22.20459	0.1900705
1965	22.2168	0.1894561
1966	22.229	0.1888439
1967	22.24121	0.1882339
1968	22.25342	0.1876261
1969	22.26563	0.1870205
1970	22.27783	0.1864171
1971	22.29004	0.1858159
1972	22.30225	0.1852168
1973	22.31445	0.1846199
1974	22.32666	0.1840252
1975	22.33887	0.1834326
1976	22.35107	0.1828422
1977	22.36328	0.1822539
1978	22.37549	0.1816677
1979	22.3877	0.1810836
1980	22.3999	0.1805016
1981	22.41211	0.1799217
1982	22.42432	0.1793439
1983	22.43652	0.1787682
1984	22.44873	0.1781945
1985	22.46094	0.1776229
1986	22.47314	0.1770533
1987	22.48535	0.1764858
1988	22.49756	0.1759203
1989	22.50977	0.1753568
1990	22.52197	0.1747954
1991	22.53418	0.1742359

原表索引	频率 (Hz)	PSD (mm²/Hz)
1992	22.54639	0.1736784
1993	22.55859	0.1731229
1994	22.5708	0.1725694
1995	22.58301	0.1720179
1996	22.59521	0.1714683
1997	22.60742	0.1709207
1998	22.61963	0.170375
1999	22.63184	0.1698313
2000	22.64404	0.1692895
2001	22.65625	0.1687496
2002	22.66846	0.1682116
2003	22.68066	0.1676756
2004	22.69287	0.1671414
2005	22.70508	0.1666091
2006	22.71729	0.1660787
2007	22.72949	0.1655502
2008	22.7417	0.1650235
2009	22.75391	0.1644987
2010	22.76611	0.1639758
2011	22.77832	0.1634546
2012	22.79053	0.1629354
2013	22.80273	0.1624179
2014	22.81494	0.1619023
2015	22.82715	0.1613884
2016	22.83936	0.1608764
2017	22.85156	0.1603662
2018	22.86377	0.1598578
2019	22.87598	0.1593511
2020	22.88818	0.1588462
2021	22.90039	0.1583431
2022	22.9126	0.1578418
2023	22.9248	0.1573422

原表索引	频率 (Hz)	PSD (mm²/Hz)
2024	22.93701	0.1568443
2025	22.94922	0.1563482
2026	22.96143	0.1558538
2027	22.97363	0.1553611
2028	22.98584	0.1548702
2029	22.99805	0.1543809
2030	23.01025	0.1538934
2031	23.02246	0.1534075
2032	23.03467	0.1529234
2033	23.04688	0.1524409
2034	23.05908	0.1519601
2035	23.07129	0.151481
2036	23.0835	0.1510035
2037	23.0957	0.1505277
2038	23.10791	0.1500535
2039	23.12012	0.149581
2040	23.13232	0.1491101
2041	23.14453	0.1486408
2042	23.15674	0.1481731
2043	23.16895	0.1477071
2044	23.18115	0.1472427
2045	23.19336	0.1467798
2046	23.20557	0.1463186
2047	23.21777	0.145859
2048	23.22998	0.1454009
2049	23.24219	0.1449444
2050	23.25439	0.1444895
2051	23.2666	0.1440361
2052	23.27881	0.1435843
2053	23.29102	0.143134
2054	23.30322	0.1426853
2055	23.31543	0.1422381

原表索引	频率 (Hz)	PSD (mm²/Hz)
2056	23.32764	0.1417925
2057	23.33984	0.1413483
2058	23.35205	0.1409057
2059	23.36426	0.1404646
2060	23.37646	0.1400251
2061	23.38867	0.139587
2062	23.40088	0.1391504
2063	23.41309	0.1387153
2064	23.42529	0.1382816
2065	23.4375	0.1378495
2066	23.44971	0.1374188
2067	23.46191	0.1369896
2068	23.47412	0.1365618
2069	23.48633	0.1361355
2070	23.49854	0.1357107
2071	23.51074	0.1352872
2072	23.52295	0.1348653
2073	23.53516	0.1344447
2074	23.54736	0.1340256
2075	23.55957	0.1336079
2076	23.57178	0.1331916
2077	23.58398	0.1327767
2078	23.59619	0.1323632
2079	23.6084	0.1319511
2080	23.62061	0.1315404
2081	23.63281	0.131131
2082	23.64502	0.1307231
2083	23.65723	0.1303165
2084	23.66943	0.1299113
2085	23.68164	0.1295075
2086	23.69385	0.129105
2087	23.70605	0.1287039

原表索引	频率 (Hz)	PSD (mm²/Hz)
2088	23.71826	0.1283041
2089	23.73047	0.1279056
2090	23.74268	0.1275085
2091	23.75488	0.1271128
2092	23.76709	0.1267183
2093	23.7793	0.1263252
2094	23.7915	0.1259334
2095	23.80371	0.1255428
2096	23.81592	0.1251536
2097	23.82813	0.1247657
2098	23.84033	0.1243791
2099	23.85254	0.1239938
2100	23.86475	0.1236098

原表索引	频率 (Hz)	PSD (mm²/Hz)
2101	23.87695	0.123227
2102	23.88916	0.1228455
2103	23.90137	0.1224653
2104	23.91357	0.1220864
2105	23.92578	0.1217087
2106	23.93799	0.1213323
2107	23.9502	0.1209571
2108	23.9624	0.1205832
2109	23.97461	0.1202105
2110	23.98682	0.119839
2111	23.99902	0.1194688
2112	24.01123	0.1190997
2113	24.02344	0.118732
2114	24.03564	0.1183654
2115	24.04785	0.118
2116	24.06006	0.1176359
2117	24.07227	0.1172729
2118	24.08447	0.1169112

原表索引	频率 (Hz)	PSD (mm²/Hz)
2119	24.09668	0.1165506
2120	24.10889	0.1161912
2121	24.12109	0.1158331
2122	24.1333	0.115476
2123	24.14551	0.1151202
2124	24.15771	0.1147655
2125	24.16992	0.114412
2126	24.18213	0.1140597
2127	24.19434	0.1137085
2128	24.20654	0.1133585
2129	24.21875	0.1130096
2130	24.23096	0.1126619
2131	24.24316	0.1123153
2132	24.25537	0.1119698
2133	24.26758	0.1116255
2134	24.27979	0.1112823
2135	24.29199	0.1109402
2136	24.3042	0.1105992
2137	24.31641	0.1102593
2138	24.32861	0.1099206
2139	24.34082	0.1095829
2140	24.35303	0.1092464
2141	24.36523	0.1089109
2142	24.37744	0.1085766
2143	24.38965	0.1082433
2144	24.40186	0.1079111
2145	24.41406	0.10758
2146	24.42627	0.10725
2147	24.43848	0.106921
2148	24.45068	0.1065931
2149	24.46289	0.1062663
2150	24.4751	0.1059405

原表索引	频率 (Hz)	PSD (mm²/Hz)
2151	24.4873	0.1056158
2152	24.49951	0.1052921
2153	24.51172	0.1049695
2154	24.52393	0.1046479
2155	24.53613	0.1043274
2156	24.54834	0.1040079
2157	24.56055	0.1036894
2158	24.57275	0.103372
2159	24.58496	0.1030556
2160	24.59717	0.1027402
2161	24.60938	0.1024258
2162	24.62158	0.1021124
2163	24.63379	0.1018001
2164	24.646	0.1014887
2165	24.6582	0.1011784
2166	24.67041	0.100869
2167	24.68262	0.1005606
2168	24.69482	0.1002533
2169	24.70703	0.09994687
2170	24.71924	0.09964147
2171	24.73145	0.09933704
2172	24.74365	0.09903359
2173	24.75586	0.0987311
2174	24.76807	0.09842959
2175	24.78027	0.09812904
2176	24.79248	0.09782945
2177	24.80469	0.09753082
2178	24.81689	0.09723314
2179	24.8291	0.09693641
2180	24.84131	0.09664063
2181	24.85352	0.09634579
2182	24.86572	0.09605188

原表索引	频率 (Hz)	PSD (mm²/Hz)
2183	24.87793	0.09575892
2184	24.89014	0.09546689
2185	24.90234	0.09517578
2186	24.91455	0.0948856
2187	24.92676	0.09459635
2188	24.93896	0.09430801
2189	24.95117	0.09402059
2190	24.96338	0.09373408
2191	24.97559	0.09344848
2192	24.98779	0.09316378
2193	25	0.09287999

输入 1：force / PSD2 / Independent force

激励 force；冻结谱单位 kN²/Hz；one-sided；Hz；linear 插值；zero 频段外；有效频段 0 — 25 Hz；总计 257 点；ZIP input-1.csv。

原表索引	频率 (Hz)	密度 (N²/Hz)
1	0	1000000
2	0.09765625	1000000
3	0.1953125	1000000
4	0.2929688	1000000
5	0.390625	1000000
6	0.4882813	1000000
7	0.5859375	1000000
8	0.6835938	1000000
9	0.78125	1000000
10	0.8789063	1000000
11	0.9765625	1000000
12	1.074219	1000000
13	1.171875	1000000
14	1.269531	1000000
15	1.367188	1000000
16	1.464844	1000000
17	1.5625	1000000

原表索引	频率 (Hz)	密度 (N²/Hz)
18	1.660156	1000000
19	1.757813	1000000
20	1.855469	1000000
21	1.953125	1000000
22	2.050781	1000000
23	2.148438	1000000
24	2.246094	1000000
25	2.34375	1000000
26	2.441406	1000000
27	2.539063	1000000
28	2.636719	1000000
29	2.734375	1000000
30	2.832031	1000000
31	2.929688	1000000
32	3.027344	1000000
33	3.125	1000000
34	3.222656	1000000
35	3.320313	1000000
36	3.417969	1000000
37	3.515625	1000000
38	3.613281	1000000
39	3.710938	1000000
40	3.808594	1000000
41	3.90625	1000000
42	4.003906	1000000
43	4.101563	1000000
44	4.199219	1000000
45	4.296875	1000000
46	4.394531	1000000
47	4.492188	1000000
48	4.589844	1000000
49	4.6875	1000000

原表索引	频率 (Hz)	密度 (N²/Hz)
50	4.785156	1000000
51	4.882813	1000000
52	4.980469	1000000
53	5.078125	1000000
54	5.175781	1000000
55	5.273438	1000000
56	5.371094	1000000
57	5.46875	1000000
58	5.566406	1000000
59	5.664063	1000000
60	5.761719	1000000
61	5.859375	1000000
62	5.957031	1000000
63	6.054688	1000000
64	6.152344	1000000
65	6.25	1000000
66	6.347656	1000000
67	6.445313	1000000
68	6.542969	1000000
69	6.640625	1000000
70	6.738281	1000000
71	6.835938	1000000
72	6.933594	1000000
73	7.03125	1000000
74	7.128906	1000000
75	7.226563	1000000
76	7.324219	1000000
77	7.421875	1000000
78	7.519531	1000000
79	7.617188	1000000
80	7.714844	1000000
81	7.8125	1000000

原表索引	频率 (Hz)	密度 (N²/Hz)
82	7.910156	1000000
83	8.007813	1000000
84	8.105469	1000000
85	8.203125	1000000
86	8.300781	1000000
87	8.398438	1000000
88	8.496094	1000000
89	8.59375	1000000
90	8.691406	1000000
91	8.789063	1000000
92	8.886719	1000000
93	8.984375	1000000
94	9.082031	1000000
95	9.179688	1000000
96	9.277344	1000000
97	9.375	1000000
98	9.472656	1000000
99	9.570313	1000000
100	9.667969	1000000
101	9.765625	1000000
102	9.863281	1000000
103	9.960938	1000000
104	10.05859	1000000
105	10.15625	1000000
106	10.25391	1000000
107	10.35156	1000000
108	10.44922	1000000
109	10.54688	1000000
110	10.64453	1000000
111	10.74219	1000000
112	10.83984	1000000
113	10.9375	1000000

原表索引	频率 (Hz)	密度 (N²/Hz)
114	11.03516	1000000
115	11.13281	1000000
116	11.23047	1000000
117	11.32813	1000000
118	11.42578	1000000
119	11.52344	1000000
120	11.62109	1000000
121	11.71875	1000000
122	11.81641	1000000
123	11.91406	1000000
124	12.01172	1000000
125	12.10938	1000000
126	12.20703	1000000
127	12.30469	1000000
128	12.40234	1000000
129	12.5	1000000
130	12.59766	1000000
131	12.69531	1000000
132	12.79297	1000000
133	12.89063	1000000
134	12.98828	1000000
135	13.08594	1000000
136	13.18359	1000000
137	13.28125	1000000
138	13.37891	1000000
139	13.47656	1000000
140	13.57422	1000000
141	13.67188	1000000
142	13.76953	1000000
143	13.86719	1000000
144	13.96484	1000000
145	14.0625	1000000

原表索引	频率 (Hz)	密度 (N²/Hz)
146	14.16016	1000000
147	14.25781	1000000
148	14.35547	1000000
149	14.45313	1000000
150	14.55078	1000000

原表索引	频率 (Hz)	密度 (N²/Hz)
151	14.64844	1000000
152	14.74609	1000000
153	14.84375	1000000
154	14.94141	1000000
155	15.03906	1000000
156	15.13672	1000000
157	15.23438	1000000
158	15.33203	1000000
159	15.42969	1000000
160	15.52734	1000000
161	15.625	1000000
162	15.72266	1000000
163	15.82031	1000000
164	15.91797	1000000
165	16.01563	1000000
166	16.11328	1000000
167	16.21094	1000000
168	16.30859	1000000
169	16.40625	1000000
170	16.50391	1000000
171	16.60156	1000000
172	16.69922	1000000
173	16.79688	1000000
174	16.89453	1000000
175	16.99219	1000000
176	17.08984	1000000

原表索引	频率 (Hz)	密度 (N²/Hz)
177	17.1875	1000000
178	17.28516	1000000
179	17.38281	1000000
180	17.48047	1000000
181	17.57813	1000000
182	17.67578	1000000
183	17.77344	1000000
184	17.87109	1000000
185	17.96875	1000000
186	18.06641	1000000
187	18.16406	1000000
188	18.26172	1000000
189	18.35938	1000000
190	18.45703	1000000
191	18.55469	1000000
192	18.65234	1000000
193	18.75	1000000
194	18.84766	1000000
195	18.94531	1000000
196	19.04297	1000000
197	19.14063	1000000
198	19.23828	1000000
199	19.33594	1000000
200	19.43359	1000000
201	19.53125	1000000
202	19.62891	1000000
203	19.72656	1000000
204	19.82422	1000000
205	19.92188	1000000
206	20.01953	1000000
207	20.11719	1000000
208	20.21484	1000000

原表索引	频率 (Hz)	密度 (N²/Hz)
209	20.3125	1000000
210	20.41016	1000000
211	20.50781	1000000
212	20.60547	1000000
213	20.70313	1000000
214	20.80078	1000000
215	20.89844	1000000
216	20.99609	1000000
217	21.09375	1000000
218	21.19141	1000000
219	21.28906	1000000
220	21.38672	1000000
221	21.48438	1000000
222	21.58203	1000000
223	21.67969	1000000
224	21.77734	1000000
225	21.875	1000000
226	21.97266	1000000
227	22.07031	1000000
228	22.16797	1000000
229	22.26563	1000000
230	22.36328	1000000
231	22.46094	1000000
232	22.55859	1000000
233	22.65625	1000000
234	22.75391	1000000
235	22.85156	1000000
236	22.94922	1000000
237	23.04688	1000000
238	23.14453	1000000
239	23.24219	1000000
240	23.33984	1000000

原表索引	频率 (Hz)	密度 (N^2/Hz)
241	23.4375	1000000
242	23.53516	1000000
243	23.63281	1000000
244	23.73047	1000000
245	23.82813	1000000
246	23.92578	1000000
247	24.02344	1000000
248	24.12109	1000000
249	24.21875	1000000
250	24.31641	1000000
251	24.41406	1000000
252	24.51172	1000000
253	24.60938	1000000
254	24.70703	1000000
255	24.80469	1000000
256	24.90234	1000000
257	25	1000000

详细原运行记录的可用性与来源

记录	来源分析 ID	可用性 / 条数
stages	analysis_d2b0b23a-117a-4217-a76d-d057b5cef5cf	已保存 / 4
mesh	analysis_c52c386c-0e16-4663-aedd-a0feac3f46b4	已保存 / 16
coordinates	analysis_c52c386c-0e16-4663-aedd-a0feac3f46b4	已保存 / 96
stiffness	analysis_c52c386c-0e16-4663-aedd-a0feac3f46b4	已保存 / 96
mass	analysis_c52c386c-0e16-4663-aedd-a0feac3f46b4	已保存 / 96

详细矩阵、坐标、网格和模态形态由数据包中冻结模态基础与 process.json 核查。矩阵行列按 coordinates 从零编号，通过 mesh 定位工程杆件与沿杆位置。K 行为力/力矩、列为位移/转角，单位 kN/m、kN/rad、kN·m/m、kN·m/rad；M 为对应单位乘 s^2。矩阵和单位广义质量振型保留内部混合单位，不套用显示长度转换。未保存的矩阵或运行阶段明确记为缺少记录；不会补算、补零或借用当前模型。